AIDCF-018-NET AI Network and Interconnect Architecture Domain Guide

AIDCF – AI Data Centre Facilities Assurance Framework

Document control

Field	Value
Document ID	AIDCF-018-NET
Title	AI Network and Interconnect Architecture Domain Guide
Version	1.0
Date	2026-10-10
Status	Draft for controlled review
Author	Siva Sethumadhavan
Parent artifact	AIDCF-018 Domain Guides
Normative controls	MCF-NET-001 to MCF-NET-024 in AIDCF-007A (control records as maintained in the canonical registry)
Depends on	AIDCF-003, AIDCF-004, AIDCF-005, AIDCF-007A, AIDCF-008, AIDCF-009, AIDCF-010, AIDCF-013, AIDCF-014
Requirement prefix	DG-NET-REQ-NNN

Revision history

Version	Date	Summary
1.0	2026-10-10	First issue of the NET domain guide, aligned to the rewritten MCF-NET control records, per-control gate mapping and canonical vocabularies.

1 Purpose and scope

This guide explains how the AI Network and Interconnect Architecture (NET) domain is assessed under AIDCF. It gives assessors, network architects, fabric operators and assurance reviewers a common technical picture of what the 24 NET controls protect, why they matter for AI capability, what evidence demonstrates each claim type, and how NET results feed critical gates and the Assurance Decision.

The guide is informative except where a statement carries a DG-NET-REQ identifier. The controls themselves (MCF-NET-001 to MCF-NET-024 in AIDCF-007A) are normative; if this guide and a control record disagree, the control record prevails and the discrepancy shall be raised as a baseline issue.

In scope:

- **Scale-out back-end fabrics** carrying accelerator-to-accelerator collective traffic, whether InfiniBand or Ethernet with RDMA over Converged Ethernet version 2 (RoCEv2) or newer Ethernet transports.
- **Scale-up domains** (intra-node and rack-scale accelerator interconnects) where they are represented as network elements; the compute aspects of scale-up integrity are assessed under CMP-02.

- **Storage networks** (dedicated fabrics, dedicated planes or traffic classes on a converged fabric).
- **Front-end networks** carrying inference requests, APIs, orchestration, data ingest and user access, and the **service gateways** on those paths.
- **Management and out-of-band (OOB) networks**, console and BMC access, and break-glass network access.
- **Physical connectivity**: structured cabling, high-density fibre, transceivers and cables, cross-connects, meet-me rooms and path records.
- **WAN and data-centre interconnect (DCI)** for multi-site training, replication, alternate capacity and carrier diversity.
- **Network control and observability**: subnet managers, SDN/EVPN controllers, configuration and firmware management, telemetry, time, DNS, DHCP, IPAM and network AAA.

Out of scope here (covered elsewhere): security policy and authority over network devices and controllers (SEC, in particular SEC-007, SEC-009 to SEC-012), accelerator and node health (CMP), storage system performance and integrity (STO), and dependency discovery as a cross-domain discipline (DEP).

2 Why this domain matters for AI capability

For conventional enterprise workloads, the network is usually a shared utility whose short disruptions are absorbed by retries. For large-scale AI training and distributed inference the network is part of the computer. Synchronous data-parallel, tensor-parallel, pipeline-parallel and expert-parallel training issues collective operations (all-reduce, all-gather, reduce-scatter, all-to-all) on every step; the step completes only when the slowest participant finishes. The consequences for assurance are:

1. **Tail behaviour, not averages, determines outcome.** A single degraded link, mis-mapped rail or congested spine port lengthens every step for every accelerator in the job. A fabric can appear healthy on average utilisation while delivering a fraction of design throughput.
2. **Failure propagates by design.** Lossless configurations (Priority-based Flow Control on Ethernet, credit-based flow control on InfiniBand) deliberately push back-pressure upstream. Correctly bounded, this prevents loss; incorrectly bounded, it spreads a local fault into a fabric-wide stall (pause storms) or a permanent stall (deadlock from cyclic buffer dependency).
3. **Common-mode exposure is high.** Back-end fabrics are built from very large numbers of identical switches, NICs and optics running the same firmware. One defective optic lot, one NIC firmware regression or one bad configuration push can affect thousands of links simultaneously (AIDCF-004 scenario X08).
4. **Recovery depends on the network first.** Restoration of storage, compute and the AI control plane presupposes working OOB management, core network, time and name resolution. A network that is the last thing to recover, or that depends on systems it is needed to recover, converts a partial outage into a prolonged one.
5. **Checkpoint traffic is a periodic stress test.** Synchronised checkpoint writes from thousands of accelerators create bursts that can saturate storage uplinks and interfere with collective traffic (scenario X06). The checkpoint path is therefore both a performance outcome and a recovery prerequisite.

NET controls therefore address performance integrity (CG-08), containment of fabric failures (CG-09), network restoration as part of recovery readiness (CG-10), and the network's contribution to dependency visibility (CG-03), change control (CG-14) and bounded digital authority (CG-15).

3 Reference patterns and variants

AIDCF does not prescribe network designs. Assessors shall nevertheless recognise the common patterns because each has characteristic failure modes and evidence.

3.1 Back-end fabric topologies

Pattern	Description	Assurance-relevant characteristics
Rail-optimised multi-plane	Ordinal NIC *n* of every node connects to leaf (rail) *n*; rails are aggregated by spines; several independent planes may exist.	Rail mapping errors force traffic across spines; plane loss reduces bandwidth proportionally; independent planes give natural fault domains if they do not share power, cooling or controllers.
Two-tier leaf-spine (folded Clos)	Every leaf connects to every spine.	Simple to reason about; bisection bandwidth set by uplink count and oversubscription; spine loss is a fractional reduction.
Three-tier fat-tree / Clos with super-spines or pods	Pods of leaf-spine joined by a super-spine or core tier.	Cross-pod traffic sees an extra tier and often oversubscription; pod boundaries are natural blast-radius boundaries; job placement matters.
Rack-scale scale-up domain	Accelerators in one or more racks joined by a dedicated high-bandwidth interconnect (switch trays or direct links).	Scale-up domain failure affects every accelerator in it; behaves as a single fault domain for many workloads; health is a CMP concern but its interconnect is a NET topology element.

3.2 Transport variants

- **InfiniBand**: credit-based link-level flow control (inherently lossless), a subnet manager computing routes and assigning addresses, partition keys for isolation, adaptive routing and, in some products, in-network reduction. Assessment focus: subnet-manager redundancy and failover at scale, partition configuration, adaptive-routing settings and management-key protection.
- **Ethernet with RoCEv2**: RDMA over UDP/IP. Commonly operated as a lossless class with PFC plus ECN-based end-to-end congestion control (DCQCN-style rate control at the NIC driven by congestion notification packets), or with lossy-tolerant transports and selective retransmission. Assessment focus: traffic-class mapping, ECN thresholds, PFC headroom and watchdog, NIC rate-control parameters and deadlock analysis.
- **Newer Ethernet AI transports** (for example those specified by the Ultra Ethernet Consortium) change congestion control and loss tolerance assumptions. The same assessment questions apply; evidence must reflect the transport actually deployed.

3.3 Other networks

- **Storage network**: dedicated fabric, dedicated plane or traffic class on the back-end or front-end fabric. Converged designs must show that checkpoint bursts do not starve collective traffic.
- **Front-end network**: leaf-spine with BGP-based routing and ECMP is typical; inference gateways, load balancers and API gateways sit on this path.
- **Management and OOB**: separate switches for BMC, switch management, PDUs and console servers, with an independent upstream (separate carrier or cellular) and local authentication fallback.
- **WAN/DCI**: dark fibre with DWDM, leased wavelengths or carrier services; diversity must be physical (routes, entries, ducts) and organisational (not two brands on one wholesale route).

3.4 Deployment variants

The AIDCF-013 reference patterns change ownership and evidence access more than technical content. In **colocation**, the operator may own cross-connects, meet-me rooms and sometimes OOB uplinks; NET controls apply to the owned portion and DEP-018 governs the boundary. In **hybrid cloud**, front-end and DCI extend into provider networks whose evidence is typically E5 provider assurance. In **shared multi-tenant platforms**, NET-012 traffic isolation becomes a primary outcome. In **brownfield retrofits**, cabling standards and path documentation are frequently the weakest controls.

4 Required functions and capability linkage

NET implements the following Required Functions in the AIDCF assurance chain. Each links to Service Outcomes defined under CAP-02 and tolerances under RES-001.

Required Function	Service Outcomes typically supported	Principal NET controls
Deliver inter-accelerator collective bandwidth and latency at required scale	Training throughput; distributed inference latency	NET-001, NET-003, NET-005, NET-007, NET-008
Contain fabric faults within declared fault domains	Training availability; multi-tenant isolation	NET-004, NET-007, NET-012, NET-021
Deliver dataset read and checkpoint write throughput	Training throughput; recovery point	NET-009
Deliver inference and API traffic with redundancy	Inference availability and latency	NET-010, NET-011
Provide independent management reach in all states	Recovery time; secure operations	NET-013 to NET-016
Maintain physical-layer integrity and traceability	All outcomes (latent common mode)	NET-002, NET-017 to NET-020
Keep network state controlled, observable and restorable	All outcomes; change integrity	NET-022, NET-023
Provide time, name, address and AAA services	Cluster bring-up; log integrity; recovery	NET-024

DG-NET-REQ-001 An assessor shall trace each NET control result used in a gate evaluation to at least one Required Function and Service Outcome of the assessed Capability Instance; a NET result with no such trace shall not be used to support a gate PASS.

5 Dependencies, fault domains and common-mode concerns

5.1 Upstream dependencies of the network

- **Power**: switches, optics and NICs draw from rack PDUs fed by the electrical distribution system. Redundant spines or planes fed from one UPS output or one switchboard section share a fault domain even if every switch has dual power supplies.
- **Cooling**: high-radix switches and high-power optics are sensitive to inlet temperature; liquid-cooled switch designs depend on the same CDU groups as compute.
- **Time, DNS, AAA**: switch configuration, telemetry and authentication depend on these core services (NET-024).
- **Controllers**: InfiniBand subnet managers, SDN/EVPN controllers and automation platforms (NET-021, NET-022).
- **Carriers and colocation operators** for WAN/DCI and cross-connects (NET-002, NET-019).

5.2 Downstream dependants

Compute job scheduling, storage access, cluster managers, monitoring, security telemetry, crisis communications and remote operations all depend on network function. NET failure therefore often appears first as a CMP, STO or OPS symptom.

5.3 Fault domains

The NET-004 fault-domain register should list at least: individual leaf, spine and super-spine switches; each rail and each plane; each pod; each subnet manager or controller instance; each OOB switch and console server; each WAN/DCI circuit and its route; and the power and cooling domains serving each of these. RES-005 integrates these into the site model; CMP-008 integrates them with compute placement.

5.4 Characteristic common modes

Common mode	Mechanism	Detection and control
Firmware/NOS version	Identical defect across all switches or NICs	Staged rollout, canary, plane-by-plane change (NET-022)
Optics lot or model	Marginal batch degrading over time	Optics inventory by lot, RMA analysis (NET-018), error trending (NET-006)
Shared lossless class	Pause propagation across tenants or across storage and collective traffic	Traffic-class design, watchdog (NET-007), isolation testing (NET-012)
Shared route or entry	Excavation, fire or flood severing nominally diverse paths	Route evidence and shared-segment register (NET-002)
Shared controller	Subnet manager or automation platform failure or compromise affecting all planes	Redundancy and placement (NET-021); security (SEC-007)
Circular service dependency	OOB authentication, DNS or AAA hosted on the infrastructure being recovered	OOB independence (NET-015); core service analysis (NET-024); RES-013

6 Threats, hazards and failure scenarios with propagation paths to Service Outcomes

The AIDCF-004 network scenario families (fabric partition, congestion collapse, scale-up link failure, management network loss, time-service failure, DNS/DHCP/IPAM failure, configuration blast radius, optical/cabling defect) and cross-domain scenarios X06, X08, X13, X15, X16 and X19 are the starting set. Representative propagation paths, expressed in the canonical chain (initiating event → path → Service Outcome effect), are below.

Scenario	Initiating event	Propagation path	Service Outcome effect
PFC pause storm	NIC firmware fault continuously asserts pause on one priority	Pause propagates leaf → spine → other leaves; lossless class stalls across pod	All jobs in pod stall; training throughput collapses until watchdog isolates the port (NET-007)
PFC deadlock	Routing change after link failure creates cyclic buffer dependency	Buffers on a cycle of switches fill and wait on each other indefinitely	Persistent stall that survives the original fault; recovery requires manual intervention
Checkpoint burst congestion (X06)	Synchronised checkpoint across thousands of ranks	Storage uplinks saturate; storage traffic shares class with collectives; head-of-line blocking	Step time spikes; checkpoint exceeds window; recovery point degraded (NET-009, RES-016)
Optic lot degradation	Batch of transceivers with rising pre-FEC errors	Corrected errors become uncorrectable; link flaps; reroute; collective timeouts	Intermittent job failures at scale; lost work beyond tolerance (NET-006, NET-018)
Firmware common mode (X08)	NOS update pushed to all spines	All planes share defect; routing or forwarding failure	Fabric-wide outage; recovery depends on rollback (NET-022)
Subnet-manager failover at scale	Master subnet manager host fails	Standby takes over but full sweep at scale exceeds collective timeout	All running jobs fail; restart from checkpoint (NET-021, NET-004)

Scenario	Initiating event	Propagation path	Service Outcome effect
Management network loss	OOB switch row on single PDU loses power during maintenance	BMC and console reach lost for rows	Diagnosis and recovery slowed; recovery time objective missed (NET-015)
Time divergence (X13)	GNSS loss with no holdover; or misconfigured time source	Clock drift; certificate validation and log correlation fail	Authentication failures; forensic and evidence integrity reduced (NET-024)
Shared authentication failure (X19)	Identity provider for network AAA hosted on the failed cluster	No administrative login to network devices; break-glass untested	Recovery blocked (NET-016, NET-024, SEC-004)
Physical intrusion or wrong-cable pull (X16)	Unauthorised or erroneous patching	Surviving path of a redundant pair severed	Loss of DCI or storage connectivity (NET-019, NET-020)
Recovery reintroduces bad configuration (X15)	Restore from backup taken after drift	Faulty congestion settings reapplied	Recurrence of instability post-recovery (NET-022, RES-015)

Threat sources relevant to NET include malicious insiders with patching access, compromised network automation, supply-chain substitution of optics or firmware, and tenants deliberately creating congestion. Authority aspects of these threats are assessed under SEC; NET assesses whether the fabric limits the consequence.

7 Safeguards

Using the AIDCF-004 safeguard functions:

- **Anticipate**: traffic-matrix analysis for oversubscription (NET-003); optics qualification (NET-018); scenario library entries for network failures (RES-003).
- **Prevent**: approved topology and reconciliation (NET-001); congestion-control design and deadlock analysis (NET-007); cabling standards (NET-017); cross-connect work control (NET-019); staged change (NET-022).
- **Detect**: latency baselines (NET-005); error and retransmission thresholds (NET-006); collective benchmarks (NET-008); telemetry coverage (NET-023); path audits (NET-020).
- **Decide and respond**: automated link isolation on error thresholds (NET-006); PFC watchdog (NET-007); controller failover (NET-021).
- **Contain**: fault-domain design (NET-004); traffic isolation (NET-012); management separation (NET-013).
- **Withstand**: path diversity (NET-002); redundant gateways (NET-011); redundant core services (NET-024).
- **Recover**: console access (NET-014); OOB resilience (NET-015); break-glass (NET-016); configuration backup and restore (NET-022).

8 Critical-gate linkage

Gate mappings are per control (SPEC §6.1). CRITICAL means the gate cannot PASS unless the control is CONFORMING at its required claim type when in scope; SUPPORTING means a NONCONFORMING result requires a documented gate-level evaluation. Meta-gates CG-11 and CG-12 are evaluated across all controls and are not mapped.

Gate	Name	CRITICAL controls	SUPPORTING controls
CG-02	Boundary and architecture integrity	—	NET-001, NET-020
CG-03	Critical dependency visibility	—	NET-002, NET-004, NET-015, NET-018, NET-020, NET-024

Gate	Name	CRITICAL controls	SUPPORTING controls
CG-08	Compute, storage and network functional integrity	NET-001, NET-007, NET-008, NET-009	NET-002, NET-003, NET-004, NET-005, NET-006, NET-010, NET-011, NET-012, NET-013, NET-017, NET-018, NET-019, NET-021, NET-022, NET-023, NET-024
CG-09	Failure containment	NET-004	NET-002, NET-007, NET-012, NET-021
CG-10	Recovery readiness	NET-015	NET-009, NET-014, NET-016, NET-020, NET-021, NET-022, NET-024
CG-14	Change control	NET-022	NET-019
CG-15	Integrated digital authority over infrastructure	—	NET-012, NET-013, NET-014, NET-015, NET-016

Rationale for the CRITICAL assignments:

- **CG-08** (functional integrity): an unreconciled topology (NET-001), untested congestion behaviour on a lossless fabric (NET-007), unmeasured collective performance (NET-008) or an unverified checkpoint path (NET-009) each directly defeats the claim that the fabric delivers the required function.
- **CG-09** (containment): NET-004 is the only control that demonstrates single-element fabric failures stay within declared blast radius.
- **CG-10** (recovery readiness): NET-015 is CRITICAL because every other restoration step depends on management reach that is independent of the failed systems.
- **CG-14** (change control): NET-022 is CRITICAL because uncontrolled NOS/firmware/configuration change across a fabric is the dominant common-mode path (X08) and cannot be compensated by site-level change governance alone.

DG-NET-REQ-002 Where a fabric is in scope for a gate to which a NET control is mapped CRITICAL, the assessor shall evaluate that control at its required claim type for that fabric; a result obtained for a different fabric of similar design shall not be substituted without a documented representativeness argument accepted by the assessment lead.

9 Capability-profile parameters

All values below are illustrative defaults for capability profiles, not requirements. A capability profile shall set site- and outcome-specific values with a recorded basis. Parameters are listed as defined in the control records.

Control	Parameter	Unit	Illustrative default	Basis
NET-001	Rail-mapping conformance	% of accelerator ports on design rail/leaf	100 (illustrative)	Rail-optimised designs assume ordinal NIC n connects to rail n; any deviation alters collective paths. Site-specific.
NET-001	Maximum topology reconciliation age	days	90 (illustrative)	Capability-profile parameter; shorten where change rate is high.
NET-002	Minimum physical separation between diverse routes	m	Site-specific; set by risk analysis (illustrative: separate containment and entry points)	Capability-profile parameter; no universal distance applies.
NET-002	Route evidence age	months	12 (illustrative)	Carriers re-groom circuits; evidence ages.

Control	Parameter	Unit	Illustrative default	Basis
NET-003	Back-end fabric design oversubscription	ratio (downlink:uplink)	1:1 (illustrative)	Common design choice for large training fabrics; site and workload specific.
NET-003	Front-end/storage uplink oversubscription	ratio	Workload-derived (illustrative 3:1)	Capability-profile parameter justified by traffic analysis.
NET-003	Minimum tolerated effective ratio under N-1	ratio	Set per Service Outcome (illustrative 1.15:1 for back-end)	Derived from performance outcome tolerance.
NET-004	Maximum endpoints affected by a single switch failure	accelerator ports	Equal to leaf downlink count (illustrative)	Derived from topology; outcome-specific tolerance.
NET-004	Maximum reroute/reconvergence time after single spine loss	s	Set by job collective timeout margin (illustrative 30)	Must be below framework collective timeouts; site-specific.
NET-004	Subnet-manager failover time	s	Vendor- and scale-dependent (illustrative 60)	Capability-profile parameter verified by test.
NET-005	Latency deviation alert threshold	% above baseline at p99	25 (illustrative)	Capability-profile parameter tuned to measurement noise.
NET-005	Baseline percentile set	percentiles	p50, p99, p99.9 (illustrative)	Tail latency drives synchronous collective step time.
NET-006	Link flap isolation threshold	flaps per hour	3 (illustrative)	Capability-profile parameter.
NET-006	Post-FEC uncorrectable codeword threshold	count per interval	Any non-zero sustained (illustrative)	Post-FEC errors become packet loss; vendor guidance applies.
NET-006	Pre-FEC BER warning threshold	BER	Vendor/PHY specific (illustrative: one order of magnitude below FEC limit)	Set from IEEE 802.3 FEC capability and vendor guidance.
NET-007	PFC watchdog detection time	ms	Vendor default unless tuned (illustrative 200)	Configured per NOS; verified by test.
NET-007	Maximum tolerated pause propagation	hops from congestion point	1 (illustrative)	Design objective to keep pause local.
NET-007	Lossless traffic classes	count	1 or 2 (illustrative)	Fewer lossless classes reduce deadlock and headroom exposure.
NET-008	Collective acceptance target	% of design bus bandwidth	90 (illustrative)	Capability-profile parameter derived from performance outcome.
NET-008	Benchmark periodic interval	days	30 (illustrative)	Detects slow drift.
NET-009	Checkpoint write window	s	Derived from checkpoint size and goodput target (illustrative 120)	Capability-profile parameter; see RES-016.
NET-009	Tolerated collective degradation during checkpoint	% bus bandwidth reduction	10 (illustrative)	Capability-profile parameter.

Control	Parameter	Unit	Illustrative default	Basis
NET-010	Front-end peak utilisation ceiling	% of uplink capacity	70 (illustrative)	Capability-profile parameter for headroom.
NET-011	Gateway capacity alert threshold	% of rated connections/throughput	70 (illustrative)	Capability-profile parameter.
NET-011	Tolerated failover disruption	s	Set by inference availability outcome (illustrative 5)	Derived from Service Outcome.
NET-012	Guaranteed bandwidth share per tenant class	% of link	Contract-specific (illustrative 25)	Set by tenant commitments (GOV-021).
NET-012	Tolerated cross-tenant degradation	% throughput reduction	5 (illustrative)	Capability-profile parameter.
NET-014	Console reachability check interval	days	30 (illustrative)	Capability-profile parameter.
NET-015	OOB power autonomy	h	Equal to or greater than black-start sequence duration (illustrative 4)	Derived from RES-014 restoration sequence.
NET-015	OOB test interval	months	6 (illustrative)	Capability-profile parameter.
NET-016	Break-glass post-use review time	h	24 (illustrative)	Capability-profile parameter.
NET-016	Break-glass test interval	months	6 (illustrative)	Capability-profile parameter.
NET-017	Maximum containment fill	% cross-section	Per adopted standard (illustrative 50)	From adopted cabling standard; site-specific.
NET-017	Fibre link loss acceptance margin	dB below transceiver budget	Set from optics budget (illustrative 1.0)	Ensures operating margin; see NET-018.
NET-018	Rx power warning margin	dB above receiver sensitivity	2 (illustrative)	Capability-profile parameter derived from vendor budget.
NET-018	Optics failure-rate review trigger	failures per 1000 per month	Site baseline-derived (illustrative)	Triggers lot-level analysis.
NET-019	Record update time after change	h	24 (illustrative)	Capability-profile parameter.
NET-020	Path record accuracy target	% of audited links correct end-to-end	99 (illustrative)	Capability-profile parameter.
NET-020	Audit sample size	links per audit	Statistically justified (illustrative 200)	Sampling per AIDCF-009.
NET-021	Controller failover time tolerance	s	Below job collective timeout margin (illustrative 60)	Capability-profile parameter.
NET-021	Controller failover test interval	months	12 (illustrative)	Capability-profile parameter.
NET-022	Drift detection interval	h	24 (illustrative)	Capability-profile parameter.
NET-022	Canary stage size	% of devices	≤5 (illustrative)	Limits firmware common-mode exposure (X08).
NET-023	Telemetry collection interval for congestion counters	s	10 (illustrative)	Capability-profile parameter; finer for diagnosis.

Control	Parameter	Unit	Illustrative default	Basis
NET-023	Telemetry retention	days	Per evidence retention class (illustrative 395)	AIDCF-008 retention classes.
NET-024	Time offset tolerance	ms	Workload- and log-correlation-dependent (illustrative 1 for NTP; sub-microsecond for PTP where required)	Capability-profile parameter.
NET-024	Holdover duration	h	Covers time-source repair time (illustrative 24)	Capability-profile parameter.

10 Evidence expectations by claim type

The NET controls use all five claim types. The following table summarises which controls carry each claim type, the evidence classes required by their profiles (AIDCF-008) and the typical decisive evidence.

Control	Claim type	Profile tier	Required classes	Quality floor	Sufficiency floor	Mandatory items
NET-003	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Per-tier ratio table; Workload-justification analysis
NET-017	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Site cabling standard
NET-001	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S4	Discovered topology export; Design-to-as-built reconciliation report
NET-002	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S4	Route evidence for each external circuit in a redundant group; Shared-segment register
NET-010	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S3	Configuration reconciliation; Attachment inventory
NET-013	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S3	Device-to-segment inventory; Reachability evidence
NET-014	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S3	Console mapping inventory; Reachability check reports
NET-018	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S3	Installed optics inventory; Qualified-optics matrix
NET-020	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S3	Audit records; Electronic reconciliation
NET-005	OPERATION	CP-3	E2+E3	P3 F3 C3 R3 T3 I3	S3	Baseline report; Probe telemetry for the period
NET-006	OPERATION	CP-3	E2+E3	P3 F3 C3 R3 T3 I3	S3	Port error telemetry; Isolation/drain ticket history
NET-011	OPERATION	CP-3	E2+E3	P3 F3 C3 R3 T3 I3	S3	Capacity telemetry; Failover evidence
NET-019	OPERATION	CP-3	E2+E3	P3 F3 C3 R3 T3 I3	S3	Work orders; Link-event logs for reconciliation
NET-022	OPERATION	CP-3	E2+E3	P3 F3 C3 R3 T3 I3	S4	Drift reports; Staged change records; Restore test record

Control	Claim type	Profile tier	Required classes	Quality floor	Sufficiency floor	Mandatory items
NET-023	OPERATION	CP-3	E2+E3	P3 F3 C3 R3 T3 I3	S3	Coverage report; Gap reports
NET-007	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Configuration reconciliation; Incast test results; Pause-storm/watchdog test result
NET-008	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Raw benchmark outputs; Approved targets
NET-009	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Checkpoint-burst test result; Traffic-class configuration extract
NET-012	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Reachability test results; Configuration-to-tenant reconciliation
NET-016	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Test records; Use and rotation records
NET-021	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Failover test record; Placement analysis
NET-024	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Circular-dependency analysis; Failure test records
NET-004	RESILIENCE	CP-5	E1+E2+E3+E4	P4 F4 C4 R4 T3 I3	S5	Fault-domain register; At least one test or analysed incident per fault-domain class; Power/cooling co-location analysis
NET-015	RESILIENCE	CP-5	E1+E2+E3+E4	P4 F4 C4 R4 T3 I3	S5	OOB dependency analysis; Isolation test record

Typical decisive evidence by claim type:

- **DESIGN** (NET-003, NET-017): approved design documents with analysis (E1 plus E7), for example a traffic-matrix analysis showing why a 1:1 back-end and 3:1 front-end ratio is adequate. Design evidence alone never supports operating or effectiveness claims.
- **IMPLEMENTATION** (NET-001, NET-002, NET-010, NET-013, NET-014, NET-018, NET-020): machine-discovered state (LLDP or subnet-manager topology, optics inventory, configuration extracts) reconciled to design. Discovery output obtained by the assessor is stronger than an operator-produced summary.
- **OPERATION** (NET-005, NET-006, NET-011, NET-019, NET-022, NET-023): telemetry and records over a representative period that includes at least one maintenance window, with exceptions and incidents reviewed.
- **EFFECTIVENESS** (NET-007, NET-008, NET-009, NET-012, NET-016, NET-021, NET-024): representative tests (incast, pause-storm, collective benchmarks, checkpoint bursts, isolation, failover) plus supporting configuration and telemetry.
- **RESILIENCE** (NET-004, NET-015): tested or analysed failure and recovery across states, with cross-domain power/cooling mapping and independent witnessing.

Evidence pitfalls specific to NET:

- **Vendor reference designs** are E1 for design intent only; they do not evidence the as-built fabric.
- **Benchmarks under non-production settings** (different collective library version, environment variables or NIC firmware) are not representative and shall be recorded as coverage gaps (U5).
- **Average utilisation dashboards** do not evidence congestion behaviour; per-class pause, ECN mark and drop counters at adequate resolution are needed.

- **Carrier diversity letters** without route detail are E1 declarations; corroborate with route maps or entry walk-downs.

11 Assessment guidance

11.1 Scoping

1. Enumerate every network supporting the Capability Instance: each back-end fabric and plane, scale-up domains, storage network, front-end, management/OOB and WAN/DCI.
2. Record ownership and evidence access per network, especially provider-operated segments.
3. Identify which networks are in scope for which gates using the mappings in Section 8.

11.2 Sequencing the assessment

A practical order is: topology and path records (NET-001, NET-020) → physical layer (NET-017, NET-018) → configuration baseline (NET-022) → telemetry coverage (NET-023) → performance and congestion (NET-005 to NET-009) → isolation and fault domains (NET-004, NET-012) → management and recovery (NET-013 to NET-016, NET-021, NET-024). Topology and configuration reconciliation come first because every later result depends on knowing what is actually deployed.

11.3 Sampling

Fabrics are large and homogeneous; sampling is acceptable for physical inspection and configuration review provided the sample is stratified by device role (leaf, spine, super-spine), plane, hardware generation and recent change. Automated full-population reconciliation is preferred where read-only access permits. Sampling rationale shall follow AIDCF-009.

11.4 Interpreting results

- A collective benchmark that meets target at full scale but fails at the declared degraded scale is PARTIAL for NET-008, not CONFORMING.
- A congestion-control design that is correct but whose configuration has drifted on a minority of switches is NONCONFORMING for NET-007 unless the drift is shown to be immaterial, because a single misconfigured switch can originate a pause storm.
- Absence of observed pause storms in telemetry is a negative claim; it requires evidence that telemetry would have detected one (resolution, coverage) before it can be relied upon.

11.5 Key assessment questions by family

- **NET-01 Scale-up and scale-out fabrics.** Does discovered topology match design on every plane, including nodes replaced since commissioning? Is each oversubscription ratio justified by the declared workload mix, including all-to-all traffic from expert-parallel models? What is the measured, not asserted, effect of losing one leaf, one spine, one plane and the primary subnet manager or controller, and do redundant fabric elements share a UPS output, switchboard section or cooling zone?
- **NET-02 Performance and congestion control.** Are latency baselines current for the deployed NOS and firmware? Are FEC, symbol-error, flap and NIC retransmission thresholds defined and acted upon? Has a pause-storm or watchdog test been run for the current NIC firmware, and has deadlock been analysed for failure routing as well as normal routing? Do collective benchmarks meet targets at the declared degraded scale?

- **NET-03 Storage and service networks.** Has a synchronised checkpoint burst been tested at declared scale with concurrent collective measurement? Are metadata services reachable over redundant paths? Are inference gateways redundant, with session behaviour on failover understood? Has RDMA-layer isolation been tested separately from IP-layer isolation on shared fabrics?
- **NET-04 Management and OOB networks.** Can every BMC, switch, PDU and storage controller be reached out of band? Does OOB depend on production identity, DNS, the production fabric or the primary carrier? Has break-glass been exercised with central authentication unreachable?
- **NET-05 Physical connectivity and optics.** Are installed optics on the qualified matrix, including module firmware? Do field labels and path records agree for an independently selected sample? Do link-event logs reconcile to approved work orders, and has anyone worked on both members of a redundant path group at once?
- **NET-06 Network control and observability.** Is NOS, NIC, optics and controller firmware held to a baseline with drift detection? Are changes staged so that no single change reaches all planes? Does telemetry resolution suffice to see congestion events? Are time, DNS, DHCP, IPAM and AAA free of circular dependencies on the cluster they serve?

DG-NET-REQ-003 An assessor shall not record NET-007 as CONFORMING for a lossless Ethernet fabric unless evidence of a pause-storm or watchdog test, or an analysed real pause-storm event, is available for the deployed NOS and NIC firmware versions.

DG-NET-REQ-004 An assessor shall record the collective library version, NIC firmware version and NOS version against every benchmark or congestion test result used as evidence, and shall treat a later change of any of these as a reassessment trigger for that evidence.

12 Testing safety

Many NET effectiveness and resilience tests are disruptive. Incast tests, pause-storm tests, switch or plane removal, subnet-manager failover, controller failover, front-end gateway failover and DCI circuit failover can each stall or terminate production workloads, and some (power-feed removal to switch rows) involve hazardous energy.

DG-NET-REQ-005 No disruptive network test shall be performed on a production fabric without written authorisation from the service owner and the network change authority, a scheduled window, workload drain or explicit workload-owner consent, a defined abort criterion and abort authority, and a rollback plan.

DG-NET-REQ-006 Tests that remove power from network equipment shall additionally be authorised under the site permit-to-work regime (OPS-007) by the electrical authority.

DG-NET-REQ-007 Assessors shall use read-only credentials and interfaces for discovery and configuration extraction; any credential with write capability issued for an assessment shall be time-bound and revoked on completion.

Preferred safe alternatives, in line with the AIDCF-004 safe validation hierarchy, are: analysis of real events with good telemetry (V2); staging fabrics at representative scale with identical software and parameters (V3/V5 with a representativeness argument); and segmented tests during planned maintenance. Scans of management networks that include OT gateways shall follow SEC-009 restrictions.

13 UNKNOWN hot-spots

The following are common sources of UNKNOWN Records in NET assessments, with typical type and severity (AIDCF-008 U-types; SPEC §5.7 severity):

Hot-spot	Typical type	Typical severity	Resolution route
As-built topology not discoverable for a fabric	U1/U5	US4 (CG-08)	Read-only discovery; witnessed port audit
NIC-side congestion parameters not extractable	U2	US4 (CG-08) where lossless	Host-side tooling under platform team; vendor support
Carrier refuses route disclosure	U2/U10	US3, US4 if sole DCI path	Contractual diversity clause; entry and first-mile survey
Benchmarks run under non-production settings	U5	US4 (CG-08)	Rerun with production environment captured
Subnet-manager failover never tested at scale	U8	US4 (CG-09/CG-08)	Staging test at scale; analysed incident
OOB dependency on production identity or DNS unanalysed	U10	US4 (CG-10)	Dependency analysis; simulated isolation test
Optics firmware not in baseline	U5	US3	Extend inventory to module firmware
Telemetry gaps over assessment period	U5	US2	Pipeline health records; alternative counter snapshots
Provider-operated front-end or cross-connects	U2	US3	E5 provider assurance; evidence rights (SUP-006)

14 Interfaces with other domains

Domain	Interface	Key controls
CMP	Rail/plane attachment, scale-up domains, NIC firmware, job placement across fault domains	CMP-005, CMP-006, CMP-008, CMP-013
STO	Checkpoint and dataset paths, metadata service reachability	STO-005, STO-008, STO-017
POW	Power paths of switches, controllers and OOB; load-step limits during restoration	POW-015, POW-024
THM	Switch cooling, liquid-cooled switch dependency on CDUs	THM-010, THM-024
DEP	Shared routes, telecom dependency, provider boundaries	DEP-007, DEP-010, DEP-018
SEC	Management segmentation, controller security, remote access, zoning	SEC-007, SEC-009, SEC-010, SEC-012
OPS	Change assessment, permit-to-work, method statements	OPS-006, OPS-007, OPS-018
RES	Site fault-domain model, recovery dependencies and sequencing	RES-005, RES-013, RES-014
SUP	Optics provenance, counterfeit prevention, firmware supply	SUP-010, SUP-014, SUP-015
PHY	Fire zones for cable routes, physical access to patching	PHY-010, PHY-013

15 Informative references

Cited at document level; no clause-level text is reproduced.

- InfiniBand Trade Association: InfiniBand Architecture Specification (Volumes 1 and 2) and the RoCEv2 annex.
- IEEE 802.1Q (bridges and bridged networks); IEEE 802.1Qbb (Priority-based Flow Control); IEEE 802.1Qaz (Enhanced Transmission Selection and DCBX).
- IEEE 802.3 (Ethernet, including optical PHYs and FEC).

- IETF RFC 3168 (Explicit Congestion Notification); RFC 7938 (BGP in large-scale data centres); RFC 5905 (NTPv4); RFC 8915 (Network Time Security); RFC 7011 (IPFIX).
- IEEE 1588 (Precision Time Protocol).
- Ultra Ethernet Consortium specification (UEC 1.0).
- OSFP MSA and QSFP-DD MSA module specifications; CMIS.
- ANSI/TIA-942; ANSI/TIA-606; ISO/IEC 11801 series; ISO/IEC 14763 series; IEC 61300-3-35; EN 50600 series / ISO/IEC 22237 series.
- OpenConfig gNMI and data models; sFlow version 5.
- ISA/IEC 62443-3-3 and NIST SP 800-82 Rev. 3 (for management-network segmentation context).
- AIDCF-004, AIDCF-005, AIDCF-008, AIDCF-009, AIDCF-010, AIDCF-013, AIDCF-014.

16 Control summary

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-NET-001	Fabric Topology	PREVENTIVE	AL3	IMPLEMENTATION	CG-08 CRITICAL; CG-02 SUPPORTING
MCF-NET-002	Path Diversity	PREVENTIVE	AL3	IMPLEMENTATION	CG-03 SUPPORTING; CG-08 SUPPORTING; CG-09 SUPPORTING
MCF-NET-003	Oversubscription	PREVENTIVE	AL2	DESIGN	CG-08 SUPPORTING
MCF-NET-004	Fabric Fault-Domain Design	PREVENTIVE	AL5	RESILIENCE	CG-09 CRITICAL; CG-08 SUPPORTING; CG-03 SUPPORTING
MCF-NET-005	Latency Baseline	DETECTIVE	AL4	OPERATION	CG-08 SUPPORTING
MCF-NET-006	Loss and Retransmission	DETECTIVE	AL4	OPERATION	CG-08 SUPPORTING
MCF-NET-007	Congestion Control	PREVENTIVE	AL5	EFFECTIVENESS	CG-08 CRITICAL; CG-09 SUPPORTING
MCF-NET-008	Collective Communication Performance	DETECTIVE	AL5	EFFECTIVENESS	CG-08 CRITICAL
MCF-NET-009	Storage Fabric	PREVENTIVE	AL5	EFFECTIVENESS	CG-08 CRITICAL; CG-10 SUPPORTING
MCF-NET-010	Frontend Network	PREVENTIVE	AL3	IMPLEMENTATION	CG-08 SUPPORTING
MCF-NET-011	Service Gateways	DETECTIVE	AL4	OPERATION	CG-08 SUPPORTING
MCF-NET-012	Traffic Isolation	PREVENTIVE	AL5	EFFECTIVENESS	CG-09 SUPPORTING; CG-08 SUPPORTING; CG-15 SUPPORTING
MCF-NET-013	Management Separation	PREVENTIVE	AL3	IMPLEMENTATION	CG-15 SUPPORTING; CG-08 SUPPORTING
MCF-NET-014	Console Access	CORRECTIVE	AL3	IMPLEMENTATION	CG-10 SUPPORTING; CG-15 SUPPORTING

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-NET-015	OOB Resilience	RECOVERY	AL5	RESILIENCE	CG-10 CRITICAL; CG-15 SUPPORTING; CG-03 SUPPORTING
MCF-NET-016	Break-Glass Access	RECOVERY	AL5	EFFECTIVENESS	CG-15 SUPPORTING; CG-10 SUPPORTING
MCF-NET-017	Cabling Standards	DIRECTIVE	AL2	DESIGN	CG-08 SUPPORTING
MCF-NET-018	Optics Compatibility	PREVENTIVE	AL3	IMPLEMENTATION	CG-08 SUPPORTING; CG-03 SUPPORTING
MCF-NET-019	Cross-Connect Control	PREVENTIVE	AL4	OPERATION	CG-14 SUPPORTING; CG-08 SUPPORTING
MCF-NET-020	Path Documentation	DETECTIVE	AL3	IMPLEMENTATION	CG-02 SUPPORTING; CG-03 SUPPORTING; CG-10 SUPPORTING
MCF-NET-021	Controller Resilience	PREVENTIVE	AL5	EFFECTIVENESS	CG-08 SUPPORTING; CG-09 SUPPORTING; CG-10 SUPPORTING
MCF-NET-022	Configuration Management	PREVENTIVE	AL4	OPERATION	CG-14 CRITICAL; CG-08 SUPPORTING; CG-10 SUPPORTING
MCF-NET-023	Flow Telemetry	DETECTIVE	AL4	OPERATION	CG-08 SUPPORTING
MCF-NET-024	Time and Core Services	PREVENTIVE	AL5	EFFECTIVENESS	CG-08 SUPPORTING; CG-03 SUPPORTING; CG-10 SUPPORTING

17 Worked synthetic example

This example is entirely synthetic. The site, figures and results are invented to illustrate the method and do not describe any real facility, organisation or test.

Context. Synthetic site S-1 hosts Capability Instance CI-T1, a training cluster of 2,048 accelerators in 256 nodes, with an eight-plane rail-optimised RoCEv2 back-end fabric, a dedicated storage plane, a leaf-spine front-end and a separate OOB network. The decision question is whether CI-T1 can sustain its training throughput outcome (≥ 85% of design step rate at full scale, ≥ 70% with one plane unavailable) and recover within a 12-hour RTO after a site-wide power event, as of the assessment date. Tolerances come from RES-001.

Selected NET results.

Control	Claim type	Evidence obtained	Result
MCF-NET-001	IMPLEMENTATION	Assessor-run LLDP discovery reconciled to design; 3 mis-mapped NIC ports on two recently replaced nodes, corrected during assessment with change records	CONFORMING (Findings closed with evidence)
MCF-NET-004	RESILIENCE	Plane-loss test witnessed; leaf-loss analysed from a real incident; subnet-manager equivalent not applicable (Ethernet); both spine groups of plane 3 found on one UPS output	PARTIAL (Finding: co-located redundancy)
MCF-NET-007	EFFECTIVENESS	Design and full configuration reconciliation; incast test passed; no pause-storm or watchdog test for current NIC firmware	UNKNOWN (U5, US4)

Control	Claim type	Evidence obtained	Result
MCF-NET-008	EFFECTIVENESS	Benchmarks at full and one-plane-down scale with production environment captured; 91% and 78% of design bus bandwidth	CONFORMING
MCF-NET-009	EFFECTIVENESS	Checkpoint burst test at full scale; checkpoint window met; collective degradation 6%	CONFORMING
MCF-NET-015	RESILIENCE	OOB design reviewed; simulated isolation test passed; OOB switches on protected supply with 6 h autonomy	CONFORMING
MCF-NET-022	OPERATION	Six months of drift reports and staged changes; one emergency NOS change applied to two planes simultaneously, retrospectively reviewed	PARTIAL (Finding)

Gate roll-up (SPEC §6.2).

- **CG-08**: CRITICAL NET controls NET-001, NET-008 and NET-009 are CONFORMING; NET-007 is UNKNOWN at US4. No CRITICAL control is NONCONFORMING or PARTIAL, so the gate is not FAIL on NET evidence; because a CRITICAL control is UNKNOWN, CG-08 = **UNKNOWN** (subject also to CMP and STO critical controls).
- **CG-09**: NET-004 is CRITICAL and PARTIAL, so CG-09 = **FAIL**. The Finding (two spine groups of one plane on one UPS output) is material because loss of that UPS output would remove an entire plane, reducing capacity to the one-plane-down level; the gate fails because PARTIAL never yields PASS for a CRITICAL control.
- **CG-10**: NET-015 is CONFORMING; the gate result depends on RES, POW, THM, CMP and STO critical controls.
- **CG-14**: NET-022 is CRITICAL and PARTIAL, so CG-14 = **FAIL** for the network scope.

Decision effect. With CG-09 and CG-14 at FAIL and CG-08 at UNKNOWN, the decision recommendation is bounded by the AIDCF-010 ceilings for those gates; the US4 UNKNOWN additionally triggers CG-12. Recommended actions: re-home one spine group of plane 3 to a different UPS output and retest plane loss; perform a watchdog test in a staging fabric with production NIC firmware or in an authorised window; amend the emergency change procedure so emergency NOS changes are applied plane-by-plane. Each action is linked to a reassessment trigger so that closure re-evaluates the affected gates rather than the score alone.

Score view. The NET domain score is computed as the mean of scored controls (CONFORMING 100, PARTIAL 50), excluding the UNKNOWN; with an UNKNOWN at US4 in scope, the band is withheld ("Band withheld (coverage/UNKNOWN)"). The score does not override the gate failures.